Prospectus:
How do context and time modulate the geometric trajectory of episodic memory?

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1 Rationale

In traditional free recall studies, participants are presented with a series of items (e.g. words, pictures, etc.), then asked to recall them to the best of their abilities. The findings yielded by such studies on recall order and transitions (Murdock, 1962; Kahana, 1996), temporal and semantic clustering tendencies (Howard and Kahana, 2002b), and response latency (Murdock and Okada, 1970) have given rise to numerous models of memory encoding, storage, and retrieval (e.g., Atkinson and Shiffrin, 1968; Howard and Kahana, 2002a; Sederberg et al., 2008). These models, in turn, have spurred investigations into the biological underpinnings of their functional predictions (e.g., Polyn et al., 2005; Manning et al., 2011). However, the free recall paradigm is limited in its ability to assess episodic memory for real-world events (see Koriat and Goldsmith, 1994; Huk et al., 2018). Its inherently discretized stimuli, binary accuracy evaluation, and reliance on participants' precisely reproduction of the presented stimuli clash with the steady, continuous evolution of real-life experiences and the value we instead place on capturing their general gist or substance in assessing our (or others') memories for them.

The emergence and popularization of naturalistic paradigms offers the opportunity to create and refine new context-driven models of episode perception and memory that better reflect the use of these systems in our day-to-day lives. In turn, such improved models better inform and constrain attempts to identify structures and networks underlying their behavioral predictions. A number of these models (e.g., Baldassano et al., 2017) have spurred imaging work that have recovered the neural representation of ongoing events at multiple, hierarchical timescales, largely based in networks of regions including the ventromedial prefrontal cortex, medial temporal lobe, and anterior cingulate cortex. These multi-timescale episodic memory findings join similar discoveries for reinforcement learning (e.g., Bernacchia et al., 2011; Bromberg-Martin et al., 2010), motor learning (e.g., Fusi et al., 2007; Kim et al., 2015), and **????** in adding weight to the long-standing challenge against the functional separation between memory and ongoing information processing (see Hasson et al., 2015). Yet, because memory for an experience is rarely as detailed as its initial perception, there must exist a mechanism for compressing and transforming experiences' content representations after they are initially perceived.

By nature, the rich complexity and fluidity of real-world experiences (as well as the numerous idiosyncratic ways they may be later described) make their content far more difficult to quantify and analyze than the simple stimuli and responses native to list-learning studies. Previous naturalistic paradigms that have attempted to investigate multi-timescale representations of episode perception and memory have had to approximate these experiences by either synthetically regulating their temporal unfolding (e.g., Lerner et al., 2011), or disregarding their content altogether and using state changes in neural responses common across subjects to infer event transitions (e.g., Howard et al., 2014).

Heusser et al. (2018a) recently developed a model that can capture the explicit moment-by-moment content of a naturalistic experience. The framework depicts the content's dynamic evolution as a geometric trajectory through a self-defined representational space. Verbal recollections of the experience may then be similarly cast as "thought trajectories" that traverse the same space. By assessing the similarity between moments of content and subsequent recountings along their trajectories' various axes, this model disentangles the overall *quality* of an episodic memory from limiting "proportion recalled" measures and enables identification of event boundaries based on the explicit content of experiences and memories.

Predictions from Heusser et al.'s (2018a) model have been successfully leveraged to implicate networks of brain regions in processing the geometric structure of an ongoing experience at multiple timescales (e.g., bilateral frontal cortex, cingulate cortex) and subsequently transforming it into compressed memory (e.g., vmPFC, ACC, rMTL), supporting and refining Baldassano et al.'s (2017) findings. Importantly, however, real-world experiences do not exist in isolation from one another. This improved, content-sensitive model enables examination of interactions between multiple uses of these processing and compression networks across time. My thesis will investigate how the geometric structure of episodic memories is modulated by factors including the passage of time, contextual congruence of existing memories, and accuracy in predicting future experiences. I hope to link my findings with established anatomical and physiological studies to inform and constrain hypotheses for investigations into these interactions' neurobiological bases.

2 Methods

2.1 Experimental design and data collection

Participants ($n = 60$) were divided into two equal groups. Both groups attended two study sessions, six to eight days apart. In the first session, all participants viewed and immediately recounted the pilot episode of *Atlanta* to the best of their abilities. Participants then predicted how the second episode will unfold the following week. In the second session, both groups began by recounting the prior week's episode. Then, group 1 viewed and recounted the second episode of *Atlanta*, while group 2 viewed and recounted the pilot episode of *Arrested Development*. All participant audio was recorded using the viewing computer's built-in microphone and automatically transcribed using Google Cloud Speech-to-Text.

2.2 Stimulus Annotation

To generate the content corpus for each stimulus video, I used `ANVIL` (Kipp, 2001), an audiovisual annotation tool, with a custom XML specification file to record the following features for each shot:

- *Narrative Details – External Events* (A few sentences describing physical occurrences and characters' actions)
- *Narrative Details – Internal State* (A few sentences describing characters' emotional states and various subtextual implications)
- A list of characters appearing on-screen
- The presence or absence of music
- A transcription of all dialogue
- The name of the character speaking
- Whether the shot took place indoors or outdoors
- The shot's setting

2.3 Dynamic content modeling, event segmentation, and video-recall mapping

Following Heusser et al. (2018a), I concatenated all featural annotations within each shot and created overlapping sliding windows of consecutive shots, upon which I trained a topic model (Blei et al., 2003) using `Hypertools` (Heusser et al., 2018b) to create a representational space based on the episode's content.

I used this model to transform the sliding windows of both the video each participant's recall and prediction transcripts into a common topic space. I then used linear interpolation to resample the video model to a one-second resolution and dynamic time warping (Berndt and Clifford, 1994) to standardize the time dimension of the recall and prediction models, separately.

I then divided the continuous video, recall, and prediction models into discrete "events." Using Hidden Markov models (Rabiner, 1989), I parsed the trajectories' timeseries into K (non-repeating) states, such that the topics represented at each moment were maximally similar within events, while maximally dissimilar across events (optimal K was computed separately for each model). To map between viewed and remembered events, I computed the average topic vector for each recalled event and and assigned it to video event whose average topic vector was maximally similar.

3 Analyses to be employed and expected results

3.1 Comparing immediate and delayed recall

To investigate how the one-week delay period alters the spatial unfolding of an episodic memory, I plan to examine differences in both the across-subjects average and individual participants' recall trajectories on a full-model scale, as well as on a per-event basis. To achieve the first, I will examine how this delay period affects the general figure of retrieved content from session 1's episode. Building on Heusser et al.'s (2018a) approach, I can use Uniform Manifold Approximation and Projection (UMAP; McInnes and Healy, 2018) to project the models down into a shared 2D plane wherein the evolving content of each stimulus traces out an unique geometric shape that is traversed by participants' 2D-embedded recall models. Heusser et al.'s embedded recall shapes showed, on average, near-perfect recovery of the video content-embedding with individuals' shapes displaying idiosyncratic differences. I plan to measure the change in the variation among individuals' recall shapes with the expectation that they will converge and "smooth" as participants retain the broad plotline of the episode and discard minor details.

Additionally, I can asses changes in both the proportion of remembered events and the quality of memory for events remembered in both recall phases. To do this, I will examine whether, upon segmenting participant's immediate and delayed recall models, the HMM identifies fewer states for the delayed model, as well as whether the topic space-distance between each remembered event and its corresponding presented event increases for events remembered in both sessions. I can then examine the features shared by video events whose memory representations were least blurred with time (i.e. those whose topic space-distance from the video trajectory increased the least) to infer features that contribute to successful maintenance in episodic memory.

3.2 Measuring the effect of contextual familiarity on future memory success

I also intend to explore how various levels of processing of prior experience modulates memory for subsequent, contextually congruent events. First, I will examine the effect mere exposure to a situation has on later encoding and retrieval of similar contexts. Most broadly, I can analyze whether participants in group 1 (i.e. those who viewed the second *Atlanta* episode during session 2) performed better than participants in group 2 (i.e. those who instead viewed *Arrested Development*) in recalling session 2's episode, by comparing metrics of proportion of events remembered and distance between video and corresponding recall models between the two groups. More finely I can also ask whether parts of session 2's episode that were highly similar to the session 1's tended to be better remembered. By projecting all three episodes' content annotations into a broadly defined topic space **should we do this with annotations of all 3 or Wikipedia content?**, I can determine level of similarity for each moment of both groups' session 2 videos to each moment of textitAtlanta episode 1 and analyze whether events in the session 2 videos were remembered more frequently or more fully as a function of that similarity.

I can then perform a similar set of analyses on the effect that the process of successfully transforming experiences into compressed memories has on subsequent iterations of that process, as well as the effect of reinstating various contextual features on subsequent memory encoding. In the case of the former, I would compute the similarities between participants' immediate recall trajectories for session 1 and session 2 in a common topic space and relate them to memory quality for session 2's episode. For the latter analysis, I would perform a similar set of operations using participants' delayed recall in place of their immediate recall.

3.3 Relating prediction ability to memory success

Similarly, I will try to determine the effect that ability to accurately predict a future experience has on its eventual successful encoding, storage, and retrieval. This may be investigated both on a full model-scale and event-wise basis. First, I will quantify overall prediction accuracy by transforming group 1 participants' prediction transcripts into the topic space defined by the content of the second *Atlanta* episode and computing the similarity between the trajectory of episode 2's content and that of each participant's prediction. I can then ask whether the similarity in overall shape between participants' memories for episode 2 and its content varies as a function of their prediction accuracy. Second, I will perform the HMM-segmentation procedure on the prediction models and determine the heavily represented topics in each prediction event. I can then ask whether memory for episode 2 events with similar topic proportions improves as a result.

4 Implications of expected results

Heusser et al. (2018a) identified a network of brain regions whose dynamic responses during stimulus viewing reflected the the temporal structure of its narrative, and a second whose dynamic responses (also during viewing) reflected the idiosyncratic temporal structure of each individual participant's yet-to-be-spoken recollections. Together, these two networks appear to process ongoing representations of experiences, and compress and transform them into memories, respectively. The function of the second network of regions (including vmPFC, ACC, and rMTL), as well as its potential impact on the first, is where I hope to direct predictions informed by my results.

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